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BC Science Connections 10

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Safety in Your Science Classroom

Safety Symbols

The following safety symbols are used in *BC Science Connections 10* to alert you to possible danger. Be sure that you understand each symbol used in an activity or investigation before you begin.



Disposal Alert

This symbol appears when care must be taken to dispose of materials properly.



Thermal Safety

This symbol appears as a reminder to be careful when handling hot objects.



Sharp Object Safety

This symbol appears when there is danger of cuts or punctures caused by the use of sharp objects.



Electrical Safety

This symbol appears as a reminder to be careful when using electrical equipment.



Skin Protection Safety

This symbol appears when the use of caustic chemicals might irritate the skin or when contact with micro-organisms might transmit infection.



Clothing Protection Safety

A lab apron should be worn when this symbol appears.



Fire Safety

This symbol appears as a reminder to be careful around open flames.



Eye Safety

This symbol appears when there is danger to the eyes and safety glasses should be worn.



Fume Safety

This symbol appears when chemicals or chemical reactions could cause dangerous fumes.



Chemical Safety

This symbol appears when chemicals could cause burns or are poisonous if absorbed through the skin.



WHMIS 2015 Symbols

Look carefully at the WHMIS (Workplace Hazardous Materials Information System) safety symbols shown here. These WHMIS symbols are used throughout Canada to identify dangerous materials. Make certain you understand what these symbols mean. When you see these symbols on containers, use safety precautions. It is also important to know that WHMIS symbols were updated in 2015. So, older textbooks or chemical containers may have the previous symbols. Those symbols are being replaced with the ones below.

Symbol	Descriptor	Meaning
	Compressed gas	For gases under pressure
	Flammable	For fire hazards
	Oxidizer	For oxidizing hazards
	Acute toxicity	Can cause death or toxicity with short exposure to small amounts
	Health hazard	May cause or be suspected of causing serious health effects
	Exclamation mark	Can cause irritation to skin and eyes
	Corrosive	For corrosive damage to metals, as well as skin and eyes
	Explosive	For explosion or reactivity hazards
	Biohazardous infectious materials	For organisms or toxins that can cause diseases in people or animals
	Environmental hazard	May cause damage to the environment



*Environmental hazards are not regulated under WHMIS, but are regulated by the Globally Harmonized System. This symbol may appear on products from other countries.

Safety Rules and Procedures

Become familiar with the following safety rules and procedures. Following them and your teacher's instructions will make performing the activities and investigations safe and enjoyable. Your teacher will also give you specific information about any other special safety rules that need to be followed in your school.

1 General rules

- Listen carefully to your teacher's instructions.
- Inform your teacher if you have any allergies, medical conditions, or other physical problems that could affect your work in the science classroom. Tell your teacher if you wear contact lenses or a hearing aid.
- Obtain your teacher's approval before beginning any activity you have designed.



- Know the location and proper use of the nearest eyewash station, deluge shower, fire extinguisher, fire blanket, first-aid kit, and fire alarm.
- Before starting an activity or investigation, read all of it. If you do not understand how to do a step, ask your teacher for help.
- Be sure you have checked the safety symbols and have read and understood the safety precautions.
- Begin an activity or investigation only after your teacher tells you to start.

2 Acting responsibly

- When you are told to do so, wear protective clothing such as a lab apron and safety goggles. Always wear protective clothing when you are using materials or equipment that may be a safety problem.
- Tie back long hair, and avoid wearing scarves, ties, or long necklaces.
- Never chew gum, eat, or drink in your science classroom. Do not taste any substance.
- Handle equipment and materials carefully. Carry only one object or container at a time.
- Inform your teacher of any spills so they can be cleaned up properly.
- Wash your hands thoroughly after doing an activity or an investigation.
- Dispose of materials as directed by your teacher.
- If other students are doing something that you consider dangerous, report it to your teacher.

3 Working with sharp objects

- Always cut away from yourself and others when using a knife or scissors.
- Always keep the pointed end of scissors or any pointed object facing away from yourself and others if you have to walk with such objects.
- If you notice sharp or jagged edges on any equipment, take special care with it and report it to your teacher.
- Dispose of broken glass according to your teacher's instructions.

4 Working with electrical equipment

- Make sure your hands are dry when touching electrical cords, plugs, or sockets.
- Pull the plug, not the cord, when unplugging electrical equipment.
- Report damaged equipment, broken ground pins, or frayed cords to your teacher.
- Place electrical cords where people will not trip over them.

5 Working with heat

- Always use heatproof containers.
- Point the open end of a container that is being heated away from yourself and others.
- Do not allow a container to boil dry.
- Handle hot objects carefully. Be especially careful with a hot plate, even if you think it has cooled down.
- If you use a laboratory burner, make sure you understand fully how to light and use it safely.
- If you do receive a burn, inform your teacher, and apply cold water to the burned area immediately.

6 Working with chemicals

- If any part of your body comes in contact with a substance, wash the area immediately and thoroughly with water. Inform your teacher.
- If you get anything in your eyes, do not touch them. Wash them in the nearest eyewash station immediately and continuously for 15 minutes, and inform your teacher.
- If you are asked to smell a substance, never smell it directly. Hold the container slightly in front of and beneath your nose, and waft the fumes towards you.
- Hold containers away from your face when pouring liquids.

7 Designing and building

- Use tools safely to cut, join, and shape objects.
- Handle modelling clay correctly. Wash your hands after using modelling clay.
- Follow proper procedures when using mechanical systems and studying their operations.
- Use special care when observing and working with objects in motion.



Welcome to Science 10 and the Process of Scientific Inquiry

Imagine taking a walk near where you live. Maybe you are familiar with the area. Maybe it's a place you don't know very well. As you observe your surroundings, what do you notice? What kinds of things do you know? What kinds of things are you curious about? What makes you wonder?

Your wonderings about the world around you are the start of a process that leads to answers. Sometimes you can find answers easily by looking online or asking a person whose opinions and knowledge you trust. Other times

you have to dig deeper by investigating and reflecting on many sources of information.

Asking questions about what you observe and wonder about is something that all scientists do. This year in science class, you will have many opportunities to observe, wonder, ask questions, and collect evidence to support the answers you develop. As you engage in this process, you will be demonstrating and developing scientific attitudes such as those shown below.

These are examples of scientific attitudes.

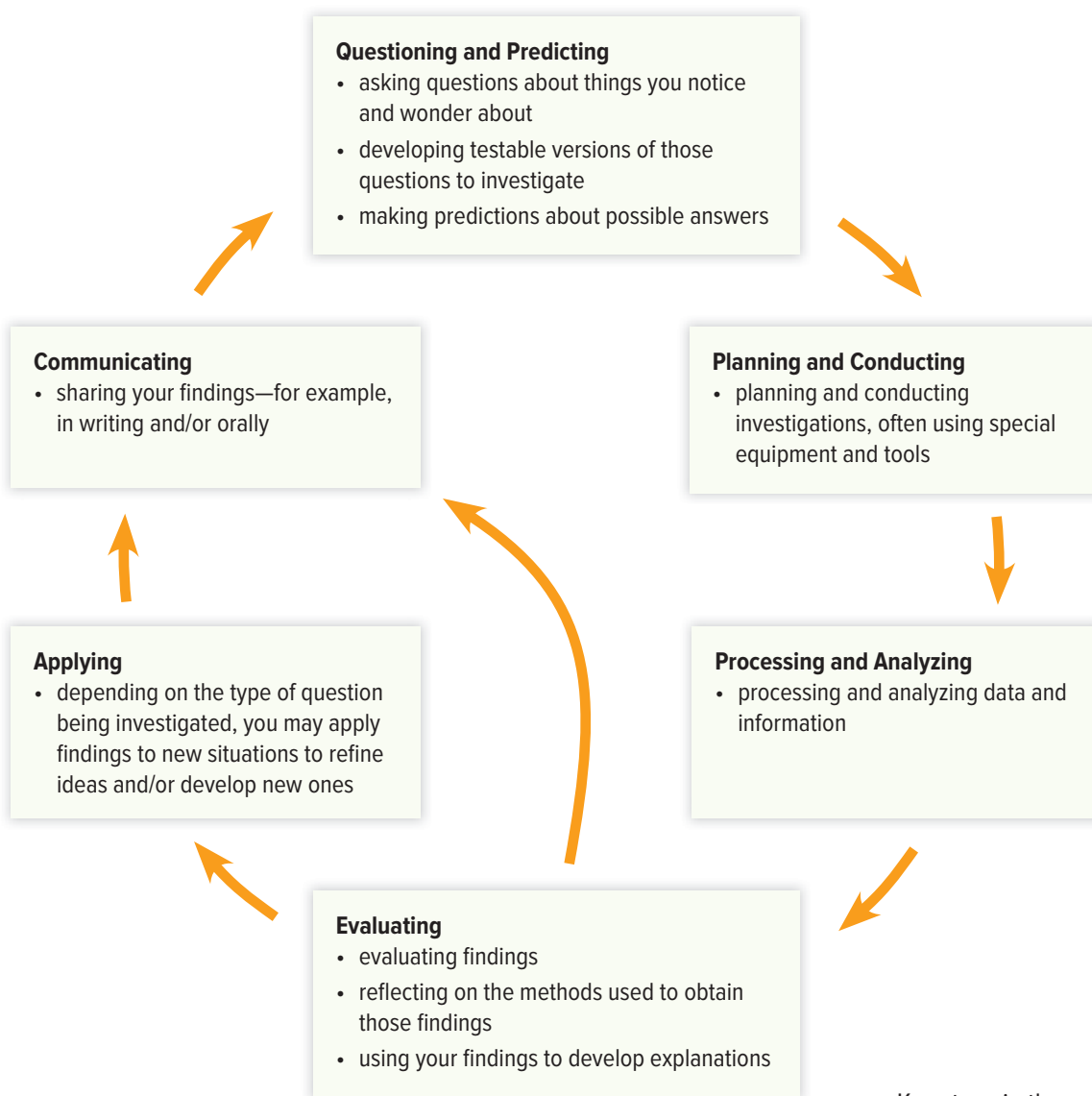
What other scientific attitudes can you think of, and what value do they bring to the work that you and other scientists do?



Scientific Inquiry

Observing, wondering, asking questions, collecting evidence, developing answers—these are all part of the process of scientific inquiry. Scientific inquiry is something that scientists do to study and propose explanations of the world, the universe. That *doing* results in a

particular kind of *knowing and understanding* of the world. Although there is no single method of scientific inquiry that everyone follows, the key steps are outlined below. Since the answers to the questions we ask can always lead to new questions, the inquiry process is often depicted in the form of a cycle diagram.



Key steps in the process of scientific inquiry

Scientific Inquiry in Your Science Class

In your class, your teacher may use the table below to help you develop your skills of scientific inquiry. If you look through this textbook, you will see that the Investigations are classified as Structured, Guided, or Open Inquiries. The first column of the table below lists the most common steps for an Investigation. In each of the next three columns, you will see the terms Teacher-led or Student-led.

Notice that in a Structured Inquiry, the teacher is responsible for the first two of the common Investigation steps, and you are responsible only for the last one. Compare the responsibilities of the teacher and students in a Structured Inquiry with the responsibilities in a Guided Inquiry and an Open Inquiry. Notice that more of the responsibilities shift from the teacher to the students. This means that as you do a variety of activities and investigations this year, you will get more practice and support to develop your inquiry skills.

Investigation Types in this Textbook

Common Investigation Steps	Structured Inquiry	Guided Inquiry	Open Inquiry
Ask a question	Teacher-led	Teacher-led	Student-led
Plan and carry out a procedure	Teacher-led	Student-led	Student-led
Analyze evidence and support an answer or claim	Student-led	Student-led	Student-led

You may be wondering what this all looks like in class. The example on the next page will help give you an idea of what you do in an inquiry investigation and some of the skills you can develop.

INVESTIGATION 2-C

STRUCTURED INQUIRY

Skills and Strategies

- Analyzing and Interpreting
- Communicating

Safety

- Wear safety goggles and lab coat or apron throughout this investigation.
- Wear gloves throughout this investigation.
- Wash any spills with plenty of water and report them to your teacher immediately.
- Dispose of materials as directed by your teacher.

What You Need

Part 1:

- 25 mL graduated cylinder
- 0.1M sodium hydroxide solution (0.1M NaOH)
- 250 mL Erlenmeyer flask
- 0.1M hydrochloric acid solution (0.1M HCl)
- Alka-Seltzer™ tablets
- 250 mL beaker
- electronic balance
- small test tube, 8 mm x 100 mm
- rubber stopper (to fit the Erlenmeyer flask)
- electronic balance (minimum readability 0.01 g)

Part 2:

1. Read Procedure steps 2 to 4. Design a table to record your results.
2. Clean up your work area and dispose of any materials according to your teacher's instructions.

Part 3:

1. Read Procedure steps 2 to 4. Design a table to record your results.
2. Clean up your work area and dispose of any materials according to your teacher's instructions.

INVESTIGATION 2-B

GUIDED INQUIRY

Skills and Strategies

- Analyzing and Interpreting
- Communicating
- Evaluating
- Planning and Designing

What You Need

- computer with internet access

The Law of Conservation of Mass

According to the law of conservation of mass, matter cannot be created or destroyed. This means that in a chemical reaction, the mass of all substances before the reaction is equal to the mass of all substances after the reaction. Using open and closed systems, you will investigate conservation of mass by analyzing the masses of substances in chemical reactions.

Question

Does the total mass of substances change in a chemical reaction?

Procedure

Part 1:

1. Read Procedure steps 2 to 4. Design a table to record your results.
2. Use a graduated cylinder to measure 20 mL of 0.1M sodium hydroxide solution, NaOH(aq). Pour the solution into a 250 mL Erlenmeyer flask.
3. Add about 1 mL of the 0.1M(1) nitric solution, HNO₃(aq), to the small test tube. This should be enough to fill the test tube about half full.
4. Fill the Erlenmeyer flask carefully and be sure the small test tube is submerged in the solution. Do not let the test tube touch the flask with the stopper.

Part 2:

1. Read Procedure steps 2 to 4. Design a table to record your results.
2. Clean up your work area and dispose of any materials according to your teacher's instructions.

Part 3:

1. Read Procedure steps 2 to 4. Design a table to record your results.
2. Clean up your work area and dispose of any materials according to your teacher's instructions.

INVESTIGATION 2-F

OPEN INQUIRY

Skills and Strategies

- Analyzing and Interpreting
- Communicating
- Evaluating
- Planning and Designing

What You Need

- computer with internet access

Chemical Reactions as Energy Sources

One way that society relies on chemical reactions is as a source of energy. Many of our most common fuels are based on chemical reactions. Some fuels are fossil fuels, such as gasoline, which are derived from ancient organic matter. Other fuels are renewable, such as hydrogen fuel cells and battery-powered cars. In this investigation, you will develop questions you have about one way our society produces energy by chemical reactions.

Procedure

1. Working in small groups, decide on what type of energy production from chemical reactions you want to investigate. This can include one of fossil fuels such as gasoline and propane at the individual level for heating, transportation, or cooking, as well as alternative energy sources such as hydrogen fuel cells.
2. Write out any questions your group has about the energy source and how chemistry is involved.
3. Decide which questions or questions you will investigate and plan how you will answer them. Your investigation should include addressing the following:
 - What is the energy used for?
 - What is the chemical reaction and balanced chemical equation that represents a description of what is occurring in the system involved and how bonds are broken and formed?
 - What are the pros and cons (including environmental and socio-economic impacts) of using this chemistry to generate the energy?

Process and Analyze

1. What is the source of and use for the energy you chose?
2. Were you able to determine the answers to all of your questions? If so, what are they? If not, why were you not able to? What information was lacking?

Conclude and Communicate

1. Summarize the advantages and disadvantages of using this chemical reaction and energy source.
2. Compare your group's findings with those of the class. In a class, generate a presentation that summarizes all of your findings.

Summary of an Inquiry: What Paper Airplane Design Will Fly the Farthest?

Inquiry Stage	What's Going On?
Making a Prediction The teacher posed a question and asked the class how to make a paper airplane fly the greatest distance. Caitlyn and Gerard were asked to make a prediction by their teacher. After some discussion, they decided that a paper airplane folded using the iconic Basic Dart design would work best.	<i>Caitlyn and Gerard are starting an inquiry investigation by making a prediction based on what they think may happen. They will test their prediction and use their data to inform their inquiry.</i>
Testing a Prediction The students were then asked to test their prediction by planning and conducting an investigation. After collecting their results, they found that their Basic Dart paper airplane had an average flight distance of 7.59 m.	<i>The students now have some data to compare against future tests.</i>
Wondering and Generating Questions Their teacher asked them to write down three things they wonder about, based on their findings. Caitlyn and Gerard agreed on the following “I wonder” statements: • Does the size of paper matter? • Are there other folding designs better than the Basic Dart? • Would changing the wing shape make a difference?	<i>The wondering questions help the students define variables that can be tested and controlled.</i>
Generating a Testable Question The students chose one wondering statement to rewrite in the form of a testable question: “Will the flight distance be longer if the shape of the wings is different?” Next, they developed a hypothesis based on their testable question: “If we fold up the edges of each wing by 2 cm, then the airplane will fly farther.” They also included a prediction based on their hypothesis: “An airplane with the edges of its wings folded up will fly farther than a flat-winged airplane.”	<i>Caitlyn and Gerard are going to change the variable of the shape of the wings and record their results. They can base their hypothesis and prediction on their results from the flat-winged paper airplane trial.</i>
Answering the Question They planned a fair test that kept everything the same as the first experiment except for the wing shape. They repeated their procedure five times and found that the design with folded-up wings had an average flight distance of 9.27 m.	<i>A fair test for any inquiry is very important. If the test isn't fair, then you can't compare the data from different trials of the investigation. Part of a fair test is ensuring that all variables but one are controlled.</i>
Analyzing Information The students analyzed their data and concluded that changing the wing shape made a difference. They communicated their findings, stating that their flat-winged paper airplane flew an average distance of 7.59 m, while the airplane with modified wings flew for an average distance of 9.27 m.	<i>Looking at their data, Caitlyn and Gerard were able to identify evidence to answer their question. Notice they used the data collected in the investigation to support their conclusion.</i>
Next Steps Afterwards, Caitlyn wondered what other modifications they could make to get even better results.	<i>Any investigation can always lead to new wonderings, new questions.</i>

First Peoples World Views in Science



Slh̓x̓7lsh or “standing man” is a landmark, near Stanley Park in Vancouver, for the Squamish, Musqueam, and Tsleil-Waututh people. The traditional narrative tells of a man who was transformed into stone as a reminder of the responsibilities of parenthood.

Science has many dimensions. Scientific knowledge, being rooted in culture, can be understood and interpreted through many different world views. In this book you are going to be asked to look at the natural world from different cultural perspectives.

The two main perspectives you will consider are often called First Peoples science and Western science. First Peoples science is the vast knowledge about the world held by First Nations, Inuit, and Métis people, based on living close to the land for thousands of years. Western science is based on ideas originally developed in Europe, with connections back to regions of the ancient Mediterranean, usually divided into specific areas of study such as biology, chemistry, and physics. While we speak of First Peoples science and Western science

as unified bodies of thought, it is important to understand that each is actually made up of a variety of different perspectives.

First Peoples scientific knowledge is based on experience gained from the local environment. Therefore, this knowledge is very diverse—as diverse as the many places that First Peoples live. However, most First Peoples share a world view that is based on a respectful and reciprocal relationship with the natural world. First Peoples science seeks to achieve a life in balance with the rest of the world.

Traditional Ecological Knowledge (TEK)

One of the most important aspects of First Peoples science is Traditional Ecological Knowledge or TEK. This is detailed local knowledge that First Peoples have gained



Ochre beds, sometimes known as the Paint Pots, in the Kootenay Mountains. In the past, First Nations used scientific knowledge to prepare the ochre by collecting the iron oxide, cleaning and baking it, then mixing with oils to create long-lasting paints.

through their relationships with the particular landscape where they live. It takes in the world view of the people. This includes relationships with their local ecosystems and the plants and animals that live there, as well as with spiritual dimensions.

The traditional knowledge of First Peoples includes understanding the lives and behaviours of plants and animals. It also includes skills and practices that people have used and developed over thousands of years to use these resources in sustainable ways. TEK embeds and embodies beliefs about the natural world and the role people play in that world. It emphasizes living in a way that respects the natural world.

For example, Traditional Ecological Knowledge includes knowing about the characteristics, behaviours, and life cycles of plants and animals. Based on their local knowledge, people are able to read signals in nature that predict the timing of events such as when a plant may ripen or salmon may arrive. This scientific knowledge is applied in many ways. In the past these ways were

crucial to the survival of the people, and today they are just as important for cultural survival. Such applications include understanding how to harvest sustainably, managing plant and animal resources by shaping the landscape, and the many ways of processing resources, such as preserving foods or tanning hides.

Traditional Ecological Knowledge is alive and ongoing. These understandings have been accumulated over many thousands of years as people interacted with the land, and they are still being gathered. Like Western science, TEK is evidence-based, collects data, and involves methods such as observation, measuring, testing, and hypothesizing.

TEK differs from Western science in significant ways, however. TEK data have been collected over thousands of years, and they have been recorded and transmitted, until very recently, only through oral traditions. As well, TEK is primarily local knowledge, learned as a result of direct contact with the local environment through observation, experience, and interactions.

A number of B.C. First Nations use the fish wheel to selectively harvest salmon in their local rivers, using modern technology to continue traditional sustainable fishing practices.





Traditional knowledge is passed on from one generation to the next through experience and story. A reciprocal relationship with the land ensures survival for future generations.

Traditional Ecological Knowledge has been passed down, generation to generation, from time immemorial, in the languages of the First Peoples who hold it. There are more than 30 different languages spoken by the First Peoples of B.C. Each has its own way of speaking about the natural world through its words and phrases, grammar, and local place names.

Knowledge and skills are passed down through story. Traditional narratives pass on information, values, and beliefs about humans' relationships with the natural world. These narratives often emphasize the need for respect by telling stories about people or creatures who did not show respect. As well as learning from story, children learn by direct experience, as they help to harvest and process the resources

from the land and participate in ceremonies and celebrations.

Today, First Peoples sometimes share their Traditional Ecological Knowledge with scientists working in fields such as wildlife and environmental sciences. Their specific knowledge of animal behaviours and their relationships with the local ecosystem are a storehouse of data that is invaluable to the work of scientists.

It is important to remember that Traditional Ecological Knowledge is the intellectual property of the people who hold it. Much of this knowledge is considered sacred and is usually not shared with outsiders. Other knowledge, such as changes in animal behaviour and properties of medicinal plants, may be shared, with the goal of contributing to wider scientific knowledge.

Renewal

Change often is cyclical—it moves in patterns that repeat themselves over and over again. Think of the seasons. Life changes over the year from young growth in the spring, ripening and maturing in the summer and autumn, and decay and sleep through the winter. But spring comes again, and life is renewed.

Renewal is key to maintaining the conditions people need to exist on the planet. But First Peoples knowledge understands that for nature to renew itself, ecosystems must be sustained. People have to act in a sustainable way to ensure the planet renews itself.

First Peoples honour the importance of renewal through important ceremonies. For example, many communities have ceremonies in the spring to mark the first berries, the first bitterroot, or the first salmon. These may be family or community events that recognize the renewal of the gifts that the natural world provides. They may include words or songs that give thanks to the plants or animals for giving up their lives to nourish humans.



Ceremonies, such as those that mark the first appearance of foods, reflect the spiritual dimension of First Peoples science. They also demonstrate the interconnectedness of people and the natural world, and they reinforce respect for resources of the land.

Reciprocity

Another theme linking all these ideas is reciprocity, which is an exchange of some kind between two systems or groups that has mutual benefit and is balanced. The natural world is built on reciprocity. You probably know the Western scientific law that for every action there is an equal and opposite reaction. This is a kind of reciprocity, but Western science tends to think of it as taking place in one direction only—from an action to a reaction, from a cause to an effect.

First Peoples science is also built on reciprocity, but it works in both directions. It is closely related to interconnectedness and interdependence. In a reciprocal relationship, there needs to be a benefit for both sides. So from a First Peoples perspective, you don't take something without giving something back.

In general terms, reciprocity is expressed as stewardship of the land. First Peoples values emphasize the responsibility of caring for the land, not just taking from it. The land gives us many gifts, which people are expected to respect and give back in return.

Often people demonstrate giving back symbolically. Many First Peoples speak words of thanks, or sing a song when they harvest plants from the land, or when they hunt an animal. Some people leave a symbolic gift such as tobacco at a harvest or sacred site.

Reciprocity is a foundation of First Peoples world view. It is evident not only in First Peoples scientific knowledge, but in social, economic, and spiritual realms. Applying reciprocity is a recognition that to maintain a sustainable universe, there needs to be a balance of give and take. Or, to put it another way, what goes around comes around.

Bridging First Peoples' Knowledge and Western Science

First Peoples perspectives on science, then, provide a holistic (all-inclusive) view of science. They seek to understand how all things are connected and dependent on each other. Western science often emphasizes the study of details, looking at certain aspects of the natural world as if they are separate from it. Sometimes it is helpful to look at the world in such detail. Neither view is better or worse than the other. We can benefit from both.

One way in which First Peoples scientific knowledge usually differs from Western science is the inclusion of spirituality. Because of their holistic view, because everything is connected, First Peoples include the spiritual side of life as part of their knowledge of the universe. It is important to note the difference between spirituality and religion. When we speak of First Peoples spiritual connections with the natural world, this is something quite different from the concepts of religious beliefs.

Spirit is that part of life that can't be observed and measured scientifically. It can be thought of as energy, or as the life force. One way to understand this is through different perspectives on animate and inanimate entities. Western science separates everything in the universe as animate or inanimate. Cougars and blueberries are animate; they are living. Rocks and stars are classified as inanimate, non-living. However, First Peoples perspectives see everything as animate, as alive. Rocks and stars have their own spirit or life force.

Western science does not usually include the spiritual side of life in its knowledge system. But there are interesting parallels with First Peoples perspectives. Consider the big bang theory of the universe. All matter is believed to have been created in the mysterious moments of the big bang. Immense forces created the stars. Almost all the elements that we know of in the universe were created and transformed in the heart of the stars. The stars exploded, sending particles of these elements through space. We are made from those particles. We are, in fact, star dust.

Pavilion Lake holds great cultural importance for the Ts'kw'aylaxw First Nation. It also has scientific importance as home to organisms linked to some of the most ancient organisms on Earth. The strong relationship built between the Ts'kw'aylaxw and scientists has resulted in valuable insights. Local knowledge leads to new discoveries that visiting scientists would not make otherwise.





These petroglyphs are found at Nanaimo in Snuneymuxw traditional territory. Such rock carvings were made at sites of power, often near water. While they have great age, these and similar rock art found around the province still hold living cultural significance for First Nations.

Another parallel between First Peoples science and Western science may be found in our dreams. For many First Peoples, dreams are an important part of understanding life. Some hunters dream about the animals they are going to hunt the next day. In the same way, many major scientific discoveries have been made through dreaming. Alfred Einstein's theory of relativity, Niels Bohr's discovery of the structure of hydrogen, and Frederick Banting's discovery of insulin all came about, in part, through dreams.

Looking at Science 10 from a First Peoples Perspective

As you study the lessons in this course, find opportunities to look at the information, concepts, and activities from a First Peoples perspective. Here are some questions you can ask to help look at the concepts in a new way:

- What can we learn from Traditional Ecological Knowledge? First Peoples have been observing and living in their local ecosystems since time immemorial. The knowledge and skills they have attained can help bring a new or additional insight to the work of Western scientists.

- How can we connect with the local First Peoples community to understand these ideas better? Can we connect with local stories and language? How can we reciprocate when they share knowledge?
- What gift from the natural world does this topic highlight? How can we give back for this gift in return?
- What is the story? Can you put the facts together to make a narrative about a topic? Does this help bring new understandings, or help you remember some ideas?
- How does this topic fit into the big picture? How can we view it holistically? Does that help us understand the problems or issues differently?
- How is this topic connected to other topics or ideas? How is it interconnected and interdependent?
- Is there evidence of transformation in this topic? Does understanding how motion and change are involved give us new ways to understand a problem?
- What is the role of renewal? Is there a question of sustainability that needs to be understood?
- How does the topic connect with you and your sense of place?

Rather than being preserved in a museum, this pole was allowed to fall and slowly return to the earth, completing the natural cycle of life.



A Tour of Your Textbook

Welcome to *BC Science Connections 10*. This textbook introduces you to the wonders of genetics and mechanisms for life's diversity, the exciting variety of chemical reactions, the amazing role of energy transformation and transfer in the world around you, and the awesome universe and how we understand it. Take a brief tour on the following pages to learn some key features of your textbook.

Unit Opener

There are four units of study in your science course—one for biology, one for chemistry, one for physics, and one for Earth and space science.

Each unit opens with a spread of two pages that feature a visual related to the unit.

The box in the upper left corner has

- the unit number
- the unit title
- an intro paragraph and quotation connected to the visual



The box in the lower right corner helps you to start thinking about the unit opener, the whole unit itself, and questions of your own that you can use to personalize your learning.



Connections to First Peoples Perspectives

This special icon identifies places in the textbook with a First Peoples context. The icon can appear on any page and as part of any component of the textbook—narrative text, activity, investigation, photo, illustration, and questions. The appearance of this icon signals an opportunity to pause and reflect on ways that First Peoples Principles of Learning may be acknowledged, understood, and respected.

Unit at a Glance

The second spread provides an overview that shows you what you will learn, do, and be responsible for.

The box in the upper left corner outlines your responsibilities—what is expected of you—for the unit.

The unit is organized into **Topics**, which are like mini-chapters. Each Topic asks a question that is a smaller, more easily explored part of the Essential Question.

Some of the things that you will do and come to know in the Topic are showcased.

The **Essential Question** is your doorway to inquiry. It echoes the Big Idea in a form that enables you to investigate important ideas and assumptions that the Big Idea represents.

UNIT 2
At a Glance

Essential Question: What happens to the energy and atoms of substances in chemical reactions?

TOPIC 2.1: How are chemical processes part of our lives?
Some things you will do:
• Gather evidence and make and carry out a test to produce a chemical reaction.
• Write a lab report and other documents to answer the question.
Some things you will come to know:
• You live in a world that is full of chemical reactions.
• You have a responsibility to use chemicals and chemical knowledge safely.

TOPIC 2.2: What happens to atoms in a chemical reaction?
Some things you will do:
• Apply First Principles perspectives and knowledge, other ways of knowing, and real knowledge to answer the question.
• Analyze, synthesize, and integrate models and diagrams.
• Write a lab report and other documents to answer the question.
Some things you will come to know:
• The number and kinds of atoms in a chemical reaction are conserved.
• Writing chemical equations depends on the law of conservation of mass.

TOPIC 2.3: How is energy involved in chemical processes?
Some things you will do:
• Collaborate and communicate your ideas, and use appropriate investigation methods.
• Analyze and synthesize data and other resources.
• Contribute to a test to see what, when, and how energy is transferred through individual and collective investigations.
Some things you will come to know:
• Chemical reactions involve a transfer of energy between systems and their surroundings.
• Some chemical reactions absorb energy and others release energy.

TOPIC 2.4: How do atoms rearrange in different types of chemical reactions?
Some things you will do:
• Use and synthesize data, models, and concepts to make a claim about a chemical reaction, and compare and contrast it to a specific process and substance.
Some things you will come to know:
• The most likely chemical reactions can be predicted using the most common chemical reactions.
• Acids and bases are common chemical substances in daily life.

Connect to What You Already Know

This feature has you reflect on prior learning to help prepare you for the unit. Additional background and reinforcement may be found in the textbook appendices and in various resources your teacher can provide.

UNIT 1
Connect To What You Already Know

This feature helps you reflect on what you know about some fundamental ideas that you have learned in previous grades. Work alone or quietly in small groups to answer the questions. Reach out to your classmates to ask about things that you are unsure about or for other assistance. Your teacher also can provide additional reinforcement materials to help you prepare for this unit.

1. Compare and contrast animal reproduction and sexual reproduction.
2. Make a diagram of a cell cycle.
3. Make a diagram of a cell cycle.
4. Explain what happens during each stage.
5. Create a series of illustrations, with captions, to describe the cell cycle.
6. What happens to DNA in each stage of the cell cycle? Why is it important that DNA is replicated during mitosis?
7. Draw a diagram of a cell with its chromosomes going through mitosis.
8. Identify whether each of the following is an event in mitosis or meiosis.
9. Homologous pairs of chromosomes meet together in the middle of the cell.
10. Crossing over occurs.
11. Homologous chromosomes move to opposite sides of the cell.

12. Explain why mitosis is related to the diversity of living things on Earth.
13. Make a diagram to show the stages involved in the formation of a gamete, beginning with one gamete from a genetic male and one gamete from a genetic female.
14. Explain the difference between haploid and diploid cells.
15. How many gametes of homologous chromosomes does a human have?
16. How many chromosomes are in each sperm cell of a male dog?
17. Which phases of mitosis are most similar to the phases of meiosis? Explain your answer.
18. Create a table to show the number of chromosomes at different stages of mitosis and meiosis.
19. Create a table to show the number of chromosomes at different stages of mitosis and meiosis.
20. Explain the difference between mitosis and meiosis.
21. Explain the difference between mitosis and meiosis.
22. Explain the difference between mitosis and meiosis.
23. Explain the difference between mitosis and meiosis.
24. Explain the difference between mitosis and meiosis.
25. Explain the difference between mitosis and meiosis.

Topic Openers

Each **Topic** opens with a spread of two pages that has the following parts:

- the Topic number and title
- an intro paragraph that works with a large photo (sometimes with a second smaller photo added)
- a list of Key Concepts and Curricular Competencies
- activities that your teacher may choose as possible Starting Points for the Topic
- a list with Key Terms that appear in the Topic

Key Concepts are the titles of the segments that make up each Topic. (See **Concepts** below.)

Curricular Competencies tell what you are expected to be able to do. Each Topic focusses on some of the many skills and processes that you will use at school, at home, and in your communities through your whole life.

TOPIC 2.4
How do atoms rearrange in different types of chemical reactions?

Key Concepts

Curricular Competencies

Starting Points

Key Terms

Concepts

The title of each **Concept** is an answer to the Topic-title question. Each Concept has some or all of the following parts.

Figures are photos, diagrams, and other visual components. Captions identify each Figure with a number and provide information about the visual. Some captions also pose questions, which appear in bold type.

Key terms are in boldface type in the narrative text. They also appear in the margin with a definition so you can find them easily. (Key terms also appear in the Glossary at the back of the book.)

CONCEPT 2 DNA technology has many uses.

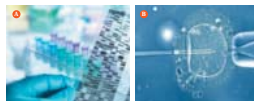
Advancements that have come from biotechnology can affect many aspects of our lives. Besides cloning and transgenic organisms, another example is a DNA profile, like the one shown in **Figure 1.37A**. A DNA profile determines the sequences of certain regions of a person's DNA that are unique to each individual. DNA profiles are used by law enforcement to try to match a person's DNA to a sample found at a crime scene. DNA profiles are also used to determine if people are related to each other.

Reproductive Technologies

Advancements in biotechnology and genetics also have touched one of the most basic biological activities—reproduction. These technologies have changed the way many people are conceiving children. They represent different options for people who want to have children but who are not able to conceive a child on their own.

In **artificial insemination**, sperm are collected and concentrated before being placed in the woman's uterus. This technique was originally developed to promote breeding success among domestic animals. Human couples have been using artificial insemination successfully for many years since it was first performed in the 1940s. Another example of the use of biotechnology to help people conceive children is **in vitro fertilization (IVF)**. In IVF, the woman's eggs are retrieved and combined with sperm in laboratory glassware. (In vitro is Latin for "in glass," and refers to the petri dish in which fertilization takes place.) After fertilization, one or more of the developing embryos are placed in the woman's uterus. Once an embryo becomes implanted, the pregnancy proceeds. In 1978, the birth of Louise Brown, the first baby born as a result of IVF, offered hope to many couples whose inability to conceive a child was due to the woman having blocked fallopian tubes. This prevents the sperm and egg from coming into contact for fertilization. A type of IVF technique, called intracytoplasmic sperm injection (ICSI) is shown in **Figure 1.37B**.

Figure 1.37 DNA profiles are used by law enforcement to identify or eliminate perpetrators of crimes. **A** is a form of IVF called intracytoplasmic sperm injection (ICSI), the sperm is injected directly into the egg.



Gene Therapy

Understanding the role genes play in human diseases is an especially important part of biotechnology. This involves finding genes or genetic material that have been mutated in some way, and then identifying a link between those changes and a particular disease. Finding the genetic material that causes a disease enables doctors to make better diagnoses and to predict the likelihood of a person developing it.

Gene therapy is an experimental treatment to cure genetic disorders by inserting a healthy, normal form of a gene into the cells of tissues that are affected by a disorder. This gene can replace the mutated gene that causes the disorder. The approach to gene therapy for trees with cystic fibrosis is shown in **Figure 1.38**. A normal gene is added to lung tissue, where it produces a normal protein. With a functioning protein, the thick mucus that builds up and harms people with cystic fibrosis is no longer produced, allowing them to breathe normally.

gene therapy an experimental treatment to cure genetic disorders that involves inserting a healthy, normal form of a gene into the cells of tissues that are affected by a disorder

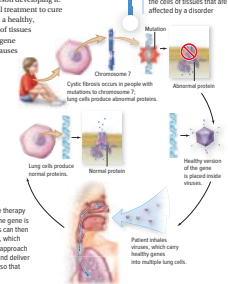


Figure 1.38 The strategy for using gene therapy to cure cystic fibrosis. A normal copy of the gene is taken up by cells in the lungs. These cells can then produce a normal protein from that gene, which can carry out its regular functions. In the approach shown, viruses act as vectors that carry and deliver the gene. The viruses have been treated so that they do not cause illness.

Extending the Connections

Gene Therapy for Donor Lungs

Scientists at the McEwen Centre for Regenerative Medicine in Toronto have developed a gene therapy technique to repair donor lungs before they are transplanted. This technique dramatically improves the success of transplants. As a result, more donor lungs are available to people who are waiting on

transplant lists. Read the information provided by your teacher about this technique. Decide, from an ethics standpoint, whether this type of gene therapy is in the same category as treating the actual genetic material of an individual. Write a paragraph that summarizes your opinion, including supporting details.

Using DNA Technology in British Columbia

The use of DNA technology can benefit British Columbia both economically and ecologically. New DNA technologies are being used to help detect viruses in plants that are imported and exported to and from Canada. As well, biotechnology is applied to help fight invasive species in provincial forests.

Food Crops and Testing for Viruses

Exporting food crops is an important source of revenue for Canada. Exported strawberries bring in about \$17 million per year, and tree fruits are worth over \$240 million per year. As you might expect, when food crops are exported and imported to and from Canada they must be tested to make sure they are not carrying any viruses that could harm other crops. If food crops, such as those shown in **Figure 1.39**, are found to have viruses during testing, they are not accepted by the buyer. The loss of sales can be costly to growers. The earlier they can detect a virus in their plants, the better. Currently, the process of testing and quarantine can take up to three years. This can affect the speed with which products are traded, which can also cost growers.



Figure 1.39 Strawberries, blueberries, and cherries are just a few of the food crops grown in and exported from British Columbia.

In 2017, the Canadian Food Inspection Agency (CFIA) announced new projects that will use biotechnology to help reduce testing and quarantine time. In one project, scientists will use DNA-based technologies to test for all viruses normally associated with imported plants. This new testing process could reduce quarantine time to only six months. The second project specifically involves the testing of strawberry plants. The process will allow scientists to carry out a single test to check for multiple viruses in all exported strawberry plants. As with the first project, this will reduce the amount of time it takes for these plants to be sold.

Invasive Species and Forests of British Columbia

British Columbia's forests are important for many reasons. People use the forests for recreation and the trees and plants in the forests reduce soil erosion, filter water, and provide habitat. Timber is an important export for British Columbia and is a \$25 billion industry across Canada.

When invasive species are introduced to forest ecosystems, they can cause great damage. An invasive species is a species that is not native to an ecosystem and causes harm to an ecosystem in which it is introduced. Examples of invasive species in B.C. forests include the Asian longhorned beetle, the Asian gypsy moth, and pathogens that cause sudden oak death and Dutch elm disease (**Figure 1.40**). Being able to identify the presence of an invasive species as early as possible can help reduce damage as well as reduce the risk that the expected timber will be replaced by the buyer.



Figure 1.40 As a caterpillar, the Asian gypsy moth eats the leaves of trees (left), bees affected by sudden oak death (middle) develop cankers on their trunks. The leaves of trees with Dutch elm disease turn yellow, wilt, and then turn brown (right).

Scientists and CFIA agents can use a handheld device to test for invasive species. The device can extract DNA from samples of insect eggs or fungi. The DNA is then compared with a library of DNA from known invasive species and pathogens. This new device can reduce the time required to identify a sample from months to days. This device can also be used to test incoming wood and plant products for the presence of invasive species or infection. Early detection can prevent invasive species from entering B.C. ecosystems and reduce the damage they cause.

Activity

Asian Gypsy Moth

Your teacher will provide you with an information sheet about the Asian gypsy moth. Read the information, and then answer the questions on the worksheet provided by your teacher.

Before you leave this page...

1. Compare and contrast artificial insemination and in vitro fertilization.
2. Choose one of the uses of biotechnology discussed in the text and describe the benefits of its use. Do you think there are any disadvantages to its use? Explain.

Extending the Connections

offer suggestions for new or additional opportunities for discovery that you can pursue on your own, with other interested classmates, and sometimes with your whole class.

An **Activity** invites you to think about or do something that is related to the Concept. Sometimes an activity may ask you to make connections to what you have learned in a previous grade or in a previous Topic.

Before you leave this page is a small set of questions to help you check your understanding before you move on.

A set of review questions appears at the end of each Topic. Different types of questions help you assess what you know and can do. The icons assigned to each question identify specific skills and processes that you demonstrate in answering the question.

Check Your Understanding of Topic 1.2

🔍 Questioning and Problems
🔍 Planning and Conducting
🔍 Processing and Analysing
🔍 Evaluating

Understand Key Ideas

- Explain how Mendel used selective breeding to learn more about heredity. 📖
- To learn of experimental design, why was it important that Mendel used true-breeding plants to explore patterns of inheritance? 📖
- Explain the differences between the following sets of terms and give an example of each:
 - dominant and recessive
 - genotype and phenotype
 - homozygous and heterozygous
- The diagram below represents the genotypes of 7 pairs of parents and one gamete from the offspring.

Connecting Ideas

- Explain how the sex of the offspring of the female parent and the genotypes of the offspring (X^h × Normal = X^hX^h + Normal = X^hX^h + Red-green colour vision deficiency = Y = Y chromosome)
 - What is the genotype of the male parent?
 - What is the phenotype of the male parent?

Making New Connections

- Sometimes breeders of plants and animals need to know if a plant or animal that has a certain phenotype has a genotype that is homozygous dominant or heterozygous. One way to determine this is by doing a test cross. At least cross involves
 - matting the individual unknown genotype with an individual who is homozygous recessive for the trait
 - analysing the phenotypes of the offspring. Figure 1.13 helps breeders identify the unknown genotype. Use a Punnett square to work through possible crosses. 📖

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TOPIC 1.2 CHECK YOUR UNDERSTANDING 39

A TOUR OF YOUR TEXTBOOK **xxxi**

Investigations

These are hands-on and minds-on opportunities to develop your science inquiry skills and to provide practice working with laboratory equipment and materials.

In a **Structured Inquiry**, you follow a specified procedure to investigate a question that is provided to you.

In a **Guided Inquiry**, you design your own procedure to investigate a question that is provided to you.

In an **Open Inquiry**, you design your own procedure to investigate a question of your own.

The image displays three sample investigation pages from the textbook, each illustrating a different type of inquiry:

- Investigation 2-C: Structured Inquiry**
 - Skills and Strategies:** Planning and Analyzing, Evaluating, Communicating.
 - Safety:** Icons for eye protection, gloves, and a spill kit.
 - What You Need:** 25 mL graduated cylinder, 100 mL water, 250 mL beaker, electronic balance, small test tubes (8 mm x 100 mm), dilute sodium hydroxide solution (0.1 mol/L NaOH(aq)), 250 mL Erlenmeyer flask, dilute iron(III) nitrate solution (0.1 mol/L Fe(NO₃)₃(aq)), Alka-Seltzer™ tablet, timer or stopwatch.
 - The Law of Conservation of Mass:** According to the law of conservation of mass, matter cannot be created or destroyed. This means that in a chemical reaction, the number of atoms of each element are the same before and after the reaction. Using open and closed systems, you will investigate the law of conservation of mass by analyzing the masses of substances in a chemical reaction.
 - Question:** Does the total mass of substances change in a chemical reaction?
 - Procedure:**
 - Read Procedure steps 2 to 7. Design a table to record your results.
 - Use a graduated cylinder to measure 20 mL of dilute sodium hydroxide solution, NaOH(aq). Pour the solution into the Erlenmeyer flask.
 - Add about 5 mL of the iron(III) nitrate solution, Fe(NO₃)₃(aq), into the small test tube. This should be enough to fill the small test tube about half full.
 - Tilt the Erlenmeyer flask carefully and let the small test tube slide down inside, as shown in the photograph. Do not let the substances mix. Seal the flask with the stopper.
- Investigation 2-B: Guided Inquiry**
 - Skills and Strategies:** Planning and Analyzing, Evaluating, Communicating.
 - What You Need:** Computer with Internet access.
 - Prepare an Emergency Response Plan:** Chemicals are part of our everyday lives but accidents can threaten the well-being of people and the environment. Being prepared for possible emergencies helps to reduce the effects of accidents. In this investigation, you will develop an emergency response plan for a hypothetical situation.
 - Question:** What features make an effective emergency response plan for a school or community?
 - Procedure:**
 - Working in small groups, decide on the type of emergency response plan you want to develop. For example, is it for a school or a small community's large city's workplace?
 - Choose one scenario for your emergency plan. Some examples are provided below.
 - A earthquake causes a gas leak in the school while school is in session.
 - An earthquake causes a gas leak in your community.
 - A transportation accident causes a dangerous chemical spill on a busy highway near a large city.
 - A fire breaks out in the school and the fire alarm goes off.
 - A fire breaks out in your home during the night.
 - Include the following considerations in your plan:
 - The preparation and planning required to be ready for an emergency. This includes the people who should be involved in organizing and executing the plan.
 - What the evacuation procedure will be and under what conditions it is to be followed.
 - How the emergency response plan will be communicated to people when the lines of communication will be among people involved in organizing and executing the plan.
 - How any contamination is to be dealt with and cleaned up; how affected material is to be disposed of.
- Investigation 2-F: Open Inquiry**
 - Skills and Strategies:** Questioning and Predicting, Planning and Analyzing, Evaluating, Communicating.
 - What You Need:** Computer with Internet access.
 - Chemical Reactions as Energy Sources:** One way that energy relies on chemical reactions is as a source of energy. Many of us use natural gas to heat our homes, gasoline to power vehicles, as well as many environmentally friendly alternatives such as hydrogen fuel cells and battery-powered cars. In this investigation, you will develop questions you have about one way we use society produces energy by chemical reactions.
 - Procedure:**
 - Working in small groups, decide on what type of energy production from chemical reactions you want to investigate. This can include use of fossil fuels such as gasoline and propane at the individual level for heating, transportation, or cooking, as well as alternative energy sources such as hydrogen fuel cells.
 - Write out any questions or topics you have about the energy source and how chemistry is involved.
 - Decide which questions or topics you will investigate and plan how you will answer them. Your investigation should include addressing the following:
 - What is the energy used for?
 - What is the chemical reaction and balance of chemical equation that represents it? Include a description of what is occurring to the atoms involved and what bonds are broken and formed.
 - What are the pros and cons (including environmental and socio-economic impacts) of using this chemistry to generate the energy?
 - Process and Analyze:**
 - What is the source of and use for the energy you chose?
 - Were you able to determine the answers to all of your questions? If so, what are they? If not, why were you not able to? What information was lacking?
 - Conclude and Communicate:**
 - Summarize the advantages and disadvantages of using this chemical reaction and energy source.
 - Combine your group's findings with those of the class. As a class, generate a presentation that summarizes all of your findings.

Pay attention to all **safety icons** and **safety warnings** so you can maintain a safe classroom environment.

Science at Work

This feature showcases a variety of jobs and careers that you might be interested to investigate on your own.

The **Science at Work: Chemistry Connections** infographic highlights various careers in chemistry and their applications:

- Forensic Scientist**
- Environmental Assessor**
- Food Scientist**
- Materials Technician**
- Laboratory Analyst**
- Mudlogger**: Ready for some rugged adventures? Mudloggers apply their knowledge of chemistry and geology as they collect and analyze rock samples from deep underground in search of oil and natural gas deposits.
- Fuel Researcher**: Interested in sustainability and a desire to reduce the effects of climate change? Fuel researchers assess the performance and emissions impact of fuels as well as investigate renewables and alternatives.
- Fire Protection Engineer**: How many jobs do you like to wear a hard hat? Fire protection engineers design, develop, and test fire protection systems to help safeguard life and property.

Questions:

- What other jobs and careers do you know or can you think of that involve working with chemicals or studying chemical reactions?
- Research a job or career related to Unit 2 that interests you. Explain what attracted you to it. What kinds of things do you have to know, do, and understand for this job or career?

This cutting-edge, multimedia-based, interactive activity enables you to demonstrate and apply knowledge, skills, and processes that you have been developing throughout the unit.



This reviews the Topic titles and Concept titles from the unit, along with Key Terms that appear in each Topic.



At the end of each unit, these pages can help you assess...

- what you know—your recall of essential information
- what you can do—your ability to engage and apply your critical and creative thinking skills
- what you understand—your ability to apply your knowledge and skills to make new connections to yourself, to society, and to the worlds that you share with all other beings

